I. a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is invertible; its inverse is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ is invertible; its inverse is $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is <u>not</u> invertible, since it is not injective.

b) Since linear transformations are in bijection with matrices (once we fix bases in the domain and codomain), it suffices to count distinct 2x2 matrices with entries in $\mathbb{Z}/5\mathbb{Z}$. So there are $5^4 = 625$.

Invertible matrices correspond to linear transformations that are isomorphisms. In particular, they map a basis to a basis. So a matrix is invertible if its columns form a basis (for the codomain). So this problem is closely related to Supp I from last week. There are 480 ordered bases for $(\mathbb{Z}/5\mathbb{Z})^2$ (240 unordered) so there are 480 invertible matrices in $M_{2\times 2}^{\mathbb{Z}/5\mathbb{Z}}$.

$$480 = \underset{\text{\# nonzero vectors in } (\mathbb{Z}/5\mathbb{Z})^2}{24} \times \underset{\text{\# vectors in } (\mathbb{Z}/5\mathbb{Z})^2 \text{ that are not scalar multiples of the first vector.}}{20}$$

II. Every subspace has a dimension, at most the dimension of the whole space, and at least 0.

There is always a unique subspace of the top dimension (the whole space) and a unique subspace of dimension 0.

How can we count subspaces for the other dimensions in between? (Assuming we are working over a finite field, so that the count is finite.)

The big idea is to overcount, and then divide away the overcounting.

$$\#(k\text{-dimensional subspaces}) = \frac{\#(\text{ordered } \cancel{\text{bases}}\ k\text{-tuples that yield a } k\text{-dimensional subspace})}{\#(\text{ordered bases that yield the } \underline{\text{same}}\ k\text{-dimensional subspace})}$$

Let's see an example.

To count 1-dimensional subspaces of $(\mathbb{Z}/5\mathbb{Z})^2$, first count how many ordered ~~bases~~ 1-tuples there are that yield 1-dimensional subspace.

Any non-zero vector v in $(\mathbb{Z}/5\mathbb{Z})^2$ is a basis for a 1-dimensional subspace. Because span$\{v\}$ is a 1-dimensional subspace, and v by itself is a basis for it.

So out numerator is $5^2 - 1 = 24$.

total vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ | the zero vector.

But we are overcounting: $\binom{1}{0}$ and $\binom{2}{0}$ and $\binom{3}{0}$ and $\binom{4}{0}$ each yield the same line (1-dim subspace).

The span of each is the same.

This same thing happens for each 1-dim subspace: it contains 4 non-zero vectors, each of which is a basis for it.

So our denominator is 4, and $\#\left(\begin{matrix}\text{1-dim subspaces}\\ \text{of } (\mathbb{Z}/5\mathbb{Z})^2\end{matrix}\right)$

$$= \frac{24}{4} = 6.$$

So $(\mathbb{Z}/5\mathbb{Z})^2$ contains $1+6+1=8$ subspaces in total.

What about $(\mathbb{Z}/5\mathbb{Z})^3$? It has 1 3-dim subspace and 1 0-dim subspace. What about 1-dim?

Again, each non-zero vector determines a 1-dim subspace, and each 1-dim subspace is determined by $5-1=4$ vectors. So we have

$$\frac{5^3-1}{4} = \frac{124}{4} = 31$$

non-zero vectors

vectors yielding the same line

And now 2-dim subspaces of $(\mathbb{Z}/5\mathbb{Z})^3$:

We want to count the number of linearly independent ordered pairs of vectors in $(\mathbb{Z}/5\mathbb{Z})^3$. Such a pair spans a 2-dim subspace, and so the pair is an ordered basis for this subspace. (linearly independent + spans).

This count is our numerator.

For the first vector, it just has to be non-zero, so $5^3-1 = 124$ choices.

The second vector in the pair has to be linearly independent from the first, and so it just has to be not a scalar multiple of the first. So 5^3-5 choices.

So our numerator is $(5^3-1)(5^3-5)$.

For the denominator: how many ordered bases does a 2-dim subspace have? Well, a 2-dim subspace is isomorphic to $(\mathbb{Z}/5\mathbb{Z})^2$! So it has $24 \times 20 = 480$ ordered bases $(=(5^2-1)(5^2-5))$

So the number of 2-dim subspaces of $(\mathbb{Z}/5\mathbb{Z})^3$ is

$$\frac{(5^3-1)(5^3-5)}{(5^2-1)(5^2-5)} = \boxed{31}$$

So the total # of subspaces of $(\mathbb{Z}/5\mathbb{Z})^3$ is $1+31+31+1 = \boxed{64}$

What about counting subspaces of $(\mathbb{Z}/p\mathbb{Z})^n$?

For the k-dimensional subspaces $(0 \le k \le n)$, we do the count in a similar fashion.

We count linearly independent ordered k-tuples, where the pool of vectors gets smaller at each stage (because we can't pick a vector that belongs to the span of the set we've built so far). So at firs the pool has $p^n - 1$ vectors (everything except the zero vector). Then everything except the vectors in the span so far: $p^n - p$, $p^n - p^2$, $\dots$, $p^n - p^{n-k-1}$

not in the 1-dim subspace so far

not in the 2-dim subspace so far

not in the $(k-1)$-dim subspace so far.

So the numerator is $(p^n - 1)(p^n - p^1)\dots(p^n - p^{n-k-1})$

The denominator is the number of ordered bases for a k-dimensional subspace, which is the same (by isomorphism!) as the number of ordered bases for $(\mathbb{Z}/p\mathbb{Z})^k$: $(p^k - 1)(p^k - p)(p^k - p^2)\dots(p^k - p^{k-1})$

So the number of k-dimensional subspaces is

$$\prod_{i=0}^{k-1} \frac{(p^n - p^i)}{(p^k - p^i)}$$

And the total number of subspaces of $(\mathbb{Z}/p\mathbb{Z})^n$ is $$\sum_{k=0}^{n} \prod_{i=0}^{k-1} \frac{p^n - p^i}{p^k - p^i}.$$

1.7.5

A subspace of $M_{4\times 4}^{\mathbb{R}}$ that contains all upper triangular matrices and all symmetric matrices must also contain any matrix that can be written as a linear combination of such matrices. (Since subspaces are closed under taking linear combinations)

But here is a basis for $M_{4\times4}^{\mathbb{R}}$ containing only upper triangular and symmetric matrices!

upper triangular

$$
\begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&1&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&1&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
$$

$$
\begin{pmatrix}0&0&0&0\\0&1&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&1&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&0&1\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&1&0\\0&0&0&0\end{pmatrix},
$$

$$
\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&1\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&1\end{pmatrix},
\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&1&0\\0&0&0&0\\1&0&0&0\\0&0&0&0\end{pmatrix},
$$

$$
\begin{pmatrix}0&0&0&1\\0&0&0&0\\0&0&0&0\\1&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&1&0\\0&1&0&0\\0&0&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&0&1\\0&0&0&0\\0&1&0&0\end{pmatrix},
\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}
$$

symmetric

So the smallest subspace containing these two subspaces is the whole space $M_{4\times4}^{\mathbb{R}}$.

1.7.5 (cont.)

The largest subspace of $M^{\mathbb{R}}_{4\times 4}$ that is contained in both of the subspaces (upper triangular and symmetric) is the subspace of diagonal matrices, that is, those of the form $\begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{pmatrix}$.

To see this, first note that by 1.7.1, the intersection of two subspaces is a subspace. So the largest possible subspace that sits inside of both upper triangular matrices and symmetric matrices is their intersection.

If a matrix is upper triangular, by definition we have that these 6 entries are 0: $\begin{pmatrix} - & - & - & - \\ 0 & - & - & - \\ 0 & 0 & - & - \\ 0 & 0 & 0 & - \end{pmatrix}$. If the matrix is also symmetric, the 6 <u>strictly</u> upper triangular entries are then also forced to be 0: $\begin{pmatrix} - & 0 & 0 & 0 \\ 0 & - & 0 & 0 \\ 0 & 0 & - & 0 \\ 0 & 0 & 0 & - \end{pmatrix}$. The remaining entries are free to be arbitrary. We see that the intersection is exactly the diagonal matrices in $M^{\mathbb{R}}_{4\times 4}$.

III. The first system is homogeneous. Its corresponding matrix (in $M_{3\times 4}^{\mathbb{Z}/5\mathbb{Z}}$) is

$$\begin{pmatrix} 2 & 3 & 1 & 1 \\ 3 & 1 & 1 & 1 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

which we now row reduce to RREF.

(This is only a solution for the first system.)

$$\xrightarrow{3R1} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 3 & 1 & 1 & 1 \\ 1 & 3 & 3 & 1 \end{pmatrix} \xrightarrow{2R2} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 1 & 2 & 2 & 2 \\ 1 & 3 & 3 & 1 \end{pmatrix} \xrightarrow[(R2+4R1)]{R2-R1} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 3 & 4 & 4 \\ 1 & 3 & 3 & 1 \end{pmatrix}$$

$$\xrightarrow{R3-R1} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 3 & 4 & 4 \\ 0 & 4 & 0 & 3 \end{pmatrix} \xrightarrow{2R2} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 3 & 3 \\ 0 & 4 & 0 & 3 \end{pmatrix} \xrightarrow{4R3} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 3 & 3 \\ 0 & 1 & 0 & 2 \end{pmatrix}$$

$$\cancel{\xrightarrow{R2\leftrightarrow R3} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 3 & 3 \end{pmatrix}} \quad \xrightarrow{R3-R2} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{3R3} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R2-3R3} \begin{pmatrix} 1 & 4 & 3 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R1-3R3} \begin{pmatrix} 1 & 4 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R1+R2} \begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$\boxed{\text{RREF}}$

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -4w \\ -2w \\ -2w \\ w \end{pmatrix} = \begin{pmatrix} 1w \\ 3w \\ 3w \\ 1w \end{pmatrix} = w \begin{pmatrix} 1 \\ 3 \\ 3 \\ 1 \end{pmatrix}$$

vector form

check:

$$\begin{pmatrix} 2 & 3 & 1 & 1 \\ 3 & 1 & 1 & 1 \\ 1 & 3 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+4+3+1 \\ 3+3+3+1 \\ 1+9+9+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

NOTE: RREF over $\mathbb{Q}$:

$$\begin{pmatrix} 1 & 0 & 0 & 1/4 \\ 0 & 1 & 0 & 1/8 \\ 0 & 0 & 1 & 1/8 \end{pmatrix}$$

V $P_3 \cong \mathbb{R}^4$

Let $\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ represent the polynomial ax^3+bx^2+cx+d.

As derivative is $3ax^2+2bx+c$, or $\begin{pmatrix} 0 \\ 3a \\ 2b \\ c \end{pmatrix}$.

If these are equal when $x=0$,

then we have $a\cdot 0^3+b\cdot 0^2+c\cdot 0+d = 3a\cdot 0^2+2b\cdot 0+c$

or $d=c$.

If these are equal when $x=1$, we have

$$a\cdot 1^3+b\cdot 1^2+c\cdot 1+d = 3a\cdot 1+2b\cdot 1+c,$$

that is, $a+b+c+d = 3a+2b+c$

that is, $2a+b-d=0$.

So the constraints on the polynomial $p(x)$ yield the system of linear equations:

$$\begin{aligned} 2a+b\quad -d &= 0 \\ c-d &= 0 \end{aligned}$$

In RREF, this is $\begin{pmatrix} 1 & \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & -1 \end{pmatrix}$.

We have
$$\begin{aligned} a &= -\tfrac{1}{2}b + \tfrac{1}{2}d \\ b &= b \\ c &= d \\ d &= d \end{aligned}$$
so the solution in vector form is:

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = b\begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ 0 \end{pmatrix} + d\begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

2.3.8.

Suppose that $Ax=0$ has a unique solution. Then the columns of A are linearly independent, since otherwise a nontrivial solution to $Ax=0$ would exist, contrary to the assumption of uniqueness. But if the columns are linearly independent, the linear transformation represented by A is injective, and therefore A has a left inverse. (As injectivity is a sufficient condition for the existence of a left inverse.)